MSA 8150 - Machine Learning for Analytics

Final Project Report

Predicting Customer Lifetime Value in Auto Insurance

Item	Detail
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Course	MSA 8150 - Machine Learning for Analytics
Institution	Georgia State University, J. Mack Robinson College of Business
Program	MS in Data Science and Analytics
Dataset	IBM Watson Marketing Customer Value Analysis
Records	9,134 customers × 24 features
Best Model	Tuned Random Forest - Test $R^2 = 0.9112$ (log scale)

Abstract

This report presents a complete supervised machine learning pipeline for predicting Customer Lifetime Value (CLV) for an auto insurance portfolio. Using 9,134 customer records from the IBM Watson Marketing Customer Value Analysis dataset, five regression models were developed, evaluated, and tuned: Linear Regression, Decision Tree, Random Forest, XGBoost, and K-Nearest Neighbors. After log-transforming the target variable to address strong right-skewness, applying domain-driven feature engineering, and conducting 5-fold cross-validated hyperparameter tuning, the Tuned Random Forest emerged as the best model with a test-set R^2 of 0.9112 on the log scale and 0.6836 in original dollar terms, with a Mean Absolute Error of $1,389 per customer. Number of Policies and Monthly Premium Auto together account for over 90% of predictive power, with direct implications for acquisition targeting, cross-sell, and retention.

1. Introduction

Customer Lifetime Value (CLV) is one of the most strategically important metrics in the insurance industry. It quantifies the total net revenue a company can expect to earn from a customer over the entire duration of their relationship. For auto insurers, accurate CLV prediction determines how much can be justified in customer acquisition costs, which segments warrant priority retention outreach, and how to allocate limited service resources across a diverse portfolio.

Without a reliable CLV model, insurers risk over-investing in low-value customers while under-retaining high-value policyholders. Traditional actuarial methods rely on population-level assumptions; machine learning captures individual non-linear patterns better suited to real portfolio complexity.

This project develops and evaluates five ML regression models for CLV prediction, with rigorous cross-validation, hyperparameter tuning, and business-level interpretation.

2. Scope and Goal of the Project

2.1 Project Scope

This project is scoped as a supervised regression task: predict each customer's CLV in dollars from observable attributes. The pipeline covers five phases:

1. Data Exploration and Cleaning - understanding distributions, feature relationships, and data quality.
2. Feature Engineering - constructing domain-informed derived features and encoding categorical variables.
3. Model Development - training and evaluating five regression models on a held-out test set.
4. Model Optimization and Validation - 5-fold cross-validation and GridSearchCV hyperparameter tuning.
5. Results Analysis and Interpretation - feature importance, residual analysis, segmented accuracy, and business implications.

2.2 Project Goal

The technical goal is the most accurate CLV regression model with sound practices - no leakage, proper train/test separation, and Pipeline-based CV. The business goal is actionable insurance strategies for acquisition targeting, cross-sell, and retention.

3. Dataset Description

3.1 Source and Context

The dataset is the IBM Watson Analytics Marketing Customer Value Analysis dataset (IBM Watson Analytics, 2019), published on Kaggle. It contains records for 9,134 auto insurance customers from five US states: Arizona, California, Nevada, Oregon, and Washington, designed for marketing analytics tasks including CLV prediction and customer segmentation.

3.2 Dataset Structure

Property	Value
Total Records	9,134 customers
Raw Features	24 columns (including Customer ID and target)
Features After ID Drop	23
Numerical Features	8 (Income, Monthly Premium Auto, CLV, Months Since Last Claim, etc.)
Categorical Features	15 (State, Vehicle Class, Coverage, Sales Channel, etc.)
Missing Values	None - dataset is complete
Target Variable	Customer Lifetime Value ($) - continuous, right-skewed
Target Range	$1,898 to $83,325
Target Mean / Median	$8,005 / $5,780

3.3 Key Features

The 23 features span four conceptual categories: policy features (Coverage, Number of Policies, Monthly Premium Auto, etc.), claims features (Total Claim Amount, Months Since Last Claim, Number of Open Complaints), demographic features (State, Gender, Marital Status, Employment Status, Income), and engagement features (Response, Sales Channel, Vehicle Class, Vehicle Size).

4. Data Exploration and Cleaning

4.1 Missing Values and Data Types

An initial audit confirmed zero missing values across all 9,134 rows - no imputation was required. The Customer ID column was dropped as a non-predictive unique identifier. Feature types were as expected: 8 numerical and 15 categorical.

4.2 CLV Distribution - Skewness Diagnosis

The distribution of Customer Lifetime Value is strongly right-skewed (skewness = 3.03). As shown in Figure 1, the majority of customers cluster below $9,000 while a long tail extends past $80,000. The mean ($8,005) substantially exceeds the median ($5,780). A log1p transformation reduces skewness to 0.58, producing a near-bell-shaped distribution suitable for regression modelling. All predictions are back-transformed using expm1() for dollar-scale reporting.

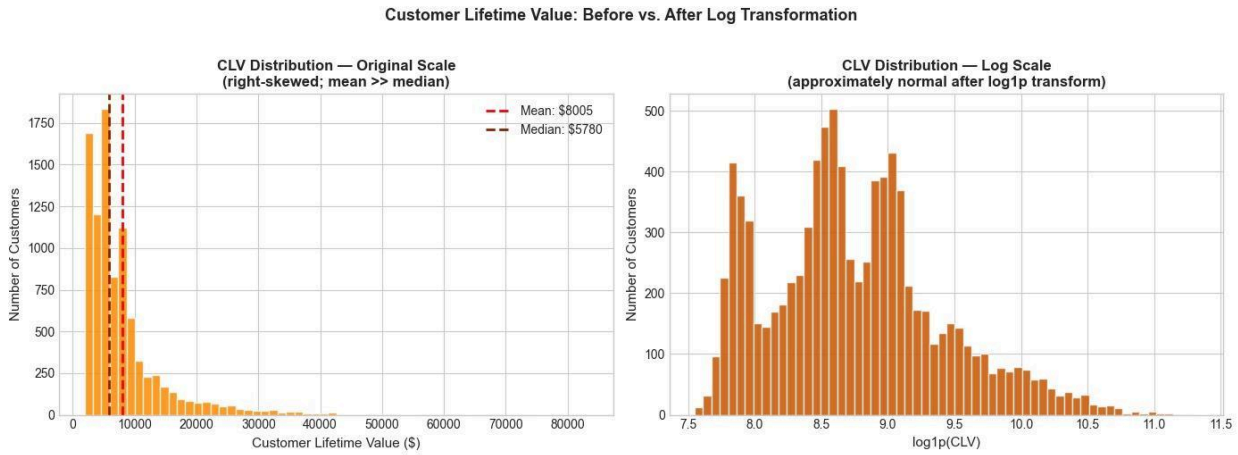


Figure 1: CLV Distribution - Original Scale (left, skewness = 3.03) and Log-transformed Scale (right, skewness = 0.58). The log1p transform stabilizes the target before modelling.

4.3 CLV by Key Categorical Features

Mean CLV was computed across three business-relevant categorical dimensions to identify natural value segments, as shown in Figure 2.

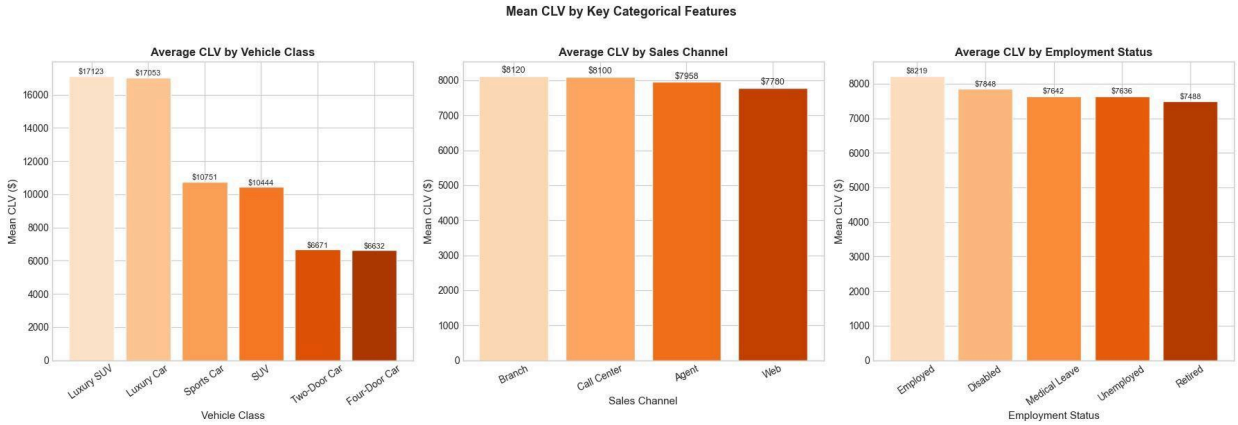


Figure 2: Mean CLV by Vehicle Class (left), Sales Channel (centre), and Employment Status (right). Luxury vehicle owners generate 2.5× the CLV of standard vehicle owners. Sales channel differences are negligible (<$400 spread).

Luxury SUV and Luxury Car owners generate ~2.5× the CLV of standard vehicles ($17,000+ vs. $6,600). Sales channel shows a spread of less than $400 across all channels and Employment Status less than $750 - neither is a meaningful CLV predictor.

4.4 Correlation Analysis - Numerical Features

Figure 3 shows the Pearson correlation heatmap. Monthly Premium Auto (r = 0.40) and Total Claim Amount (r = 0.23) are the only meaningful linear predictors; all others are near-zero, confirming dominant non-linear relationships that motivate ensemble models.

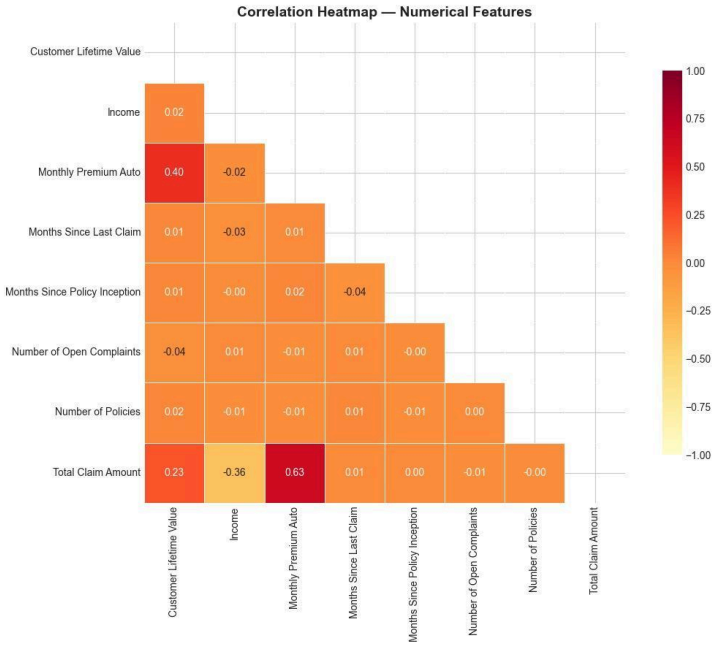


Figure 3: Correlation Heatmap - Numerical Features. Monthly Premium Auto (r = 0.40) and Total Claim Amount (r = 0.23) are the only features with meaningful linear correlation with CLV.

Feature	Pearson r with CLV	Interpretation
Monthly Premium Auto	0.396	Strongest linear signal
Total Claim Amount	0.226	Moderate positive correlation
Income	0.024	Near-zero linear relationship
Number of Policies	0.022	Near-zero linear, but non-linearly dominant in tree models
Months Since Last Claim	0.012	Minimal relationship
Months Since Policy Inception	0.009	Minimal relationship
Number of Open Complaints	-0.036	Slight negative association

4.5 Zero-Income Customers

Approximately 25.4% of customers (2,317 records) report zero income. Domain review confirmed these are not errors - they represent unemployed, disabled, or retired customers with active policies. Treating these as missing would introduce bias; they were retained as valid observations.

5. Feature Engineering

5.1 Target Transformation

The target variable was log1p-transformed (log(CLV + 1)) to produce log_CLV, reducing skewness from 3.03 to 0.58. Without this step, models are disproportionately penalized for errors on high-CLV outliers, biasing predictions toward the distribution center.

5.2 Derived Numerical Features

Feature Name	Formula	Rationale
Premium_to_Income_Ratio	Monthly Premium / (Income + 1)	Affordability burden; +1 prevents division by zero for zero-income customers
Claim_per_Policy	Total Claim Amount / Number of Policies	Per-policy risk exposure - distributes total claims across all policies held
High_Complaint_Flag	1 if Open Complaints > 0, else 0	Binary dissatisfaction indicator; correlated with churn risk
Policy_Tenure_Bucket	Quartile bin of Months Since Policy Inception	Captures non-linear tenure effects in ordinal categorical form
Renewal_Month	Calendar month from Effective To Date	Captures seasonality in renewal behavior and policy cycle timing

5.3 Categorical Encoding and Scaling

All 15 categorical features were one-hot encoded using pd.get_dummies() with drop_first=True to avoid perfect multicollinearity, expanding the feature space from 23 to 56 columns. Numerical features were standardized using StandardScaler, fitted exclusively on the training set and applied inside Pipeline objects during cross-validation to prevent any data leakage across folds.

6. Model Development

6.1 Train-Test Split

The dataset was split 80/20 (train: 7,307; test: 1,827; random_state=42), with the test set strictly withheld from all training and tuning.

6.2 Models Selected and Justification

Model	Type	Key Justification
Linear Regression	Linear baseline	Establishes a baseline; tests whether CLV relationships are linear
Decision Tree	Tree-based, single	Captures non-linearities and interactions; interpretable structure
Random Forest	Ensemble (Bagging)	Reduces Decision Tree variance by averaging many trees; state-of-the-art for tabular data
XGBoost	Ensemble (Boosting)	Sequentially corrects residuals with regularization; competitive with Random Forest
KNN	Instance-based	Non-parametric sanity check; sensitive to high dimensionality

6.3 Test Set Results

Model	RMSE	MAE	R^2	Rank
Random Forest	0.1982	0.0911	0.9113	1st
XGBoost	0.2079	0.1088	0.9023	2nd
Decision Tree	0.2189	0.1017	0.8918	3rd
Linear Regression	0.5027	0.3811	0.4291	4th
KNN	0.5877	0.4384	0.2196	5th

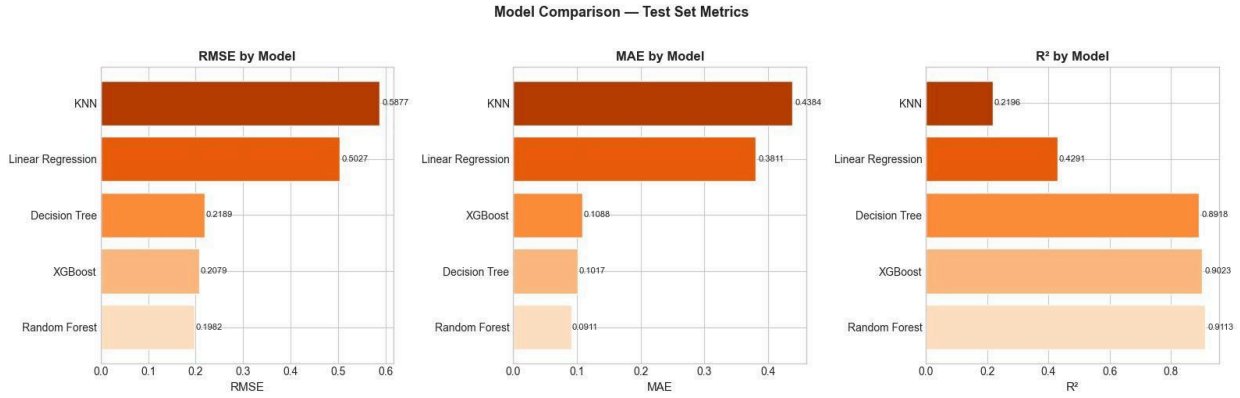


Figure 4: Model Comparison - Test Set RMSE, MAE, and R² (log_CLV scale). Random Forest achieves the lowest RMSE (0.1982) and highest R² (0.9113).

Ensemble tree models dominate; the R² gap (0.91 vs. 0.43) confirms predominantly non-linear CLV relationships. KNN performs worst due to the curse of dimensionality in the 56-feature space.

7. Model Optimization and Validation

7.1 5-Fold Cross-Validation

Each model was wrapped in a Pipeline with StandardScaler and evaluated using 5-fold KFold CV on the training set only, confirming generalization across data subsets.

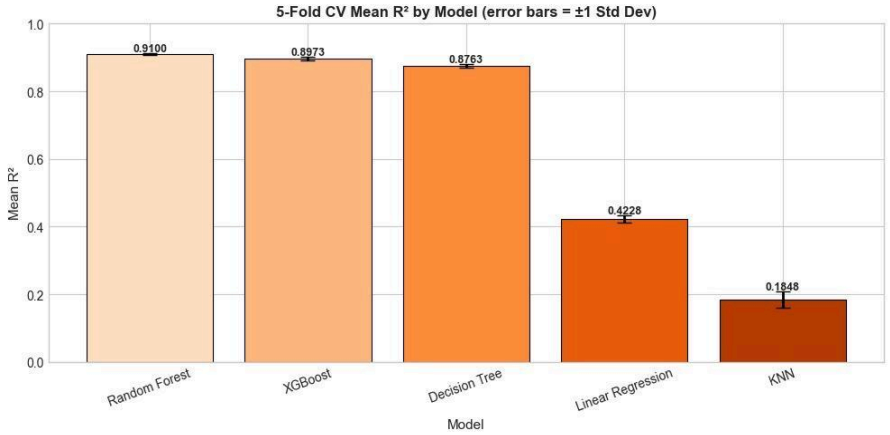


Figure 5: 5-Fold CV Mean R² ± 1 Std Dev by Model. Random Forest achieves the highest mean R² (0.9100) with the smallest variance (±0.0036), confirming it is the most stable model.

Model	CV Mean R²	CV Std Dev	Interpretation
Random Forest	0.9100	±0.0036	Highest R², narrowest variance - most stable
XGBoost	0.8973	±0.0053	Excellent and very stable - close second
Decision Tree	0.8763	±0.0053	Good but slightly below ensembles
Linear Regression	0.4228	±0.0116	Captures partial signal only
KNN	0.1848	±0.0247	Poorest; highest fold-to-fold variance

7.2 GridSearchCV Hyperparameter Tuning

The two best models were refined using 5-fold GridSearchCV on the training set only.

Random Forest (60 fits): Best params - n_estimators=200, max_depth=None, min_samples_split=5

Result: Best CV R² = 0.9095 | Test RMSE = 0.1983 | Test R² = 0.9112

XGBoost (90 fits): Best params - n_estimators=100, max_depth=5, learning_rate=0.1
Result: Best CV $R^2 = 0.9073$ | Test RMSE = 0.2022 | Test $R^2 = 0.9076$

The Tuned Random Forest ($R^2 = 0.9112$) outperforms Tuned XGBoost ($R^2 = 0.9076$) on both metrics and is selected as the final model.

8. Results Analysis and Interpretation

8.1 Final Model Performance

Metric	Log Scale	Original Dollar Scale
R^2	0.9112	0.6836
RMSE	0.1983 log units	$4,037.99
MAE	0.0911 log units	$1,389.21
Mean Residual	-	+$353.30 (small under-prediction bias)
Mean Actual CLV (test)	-	$7,933.55
Std Dev Actual CLV	-	$7,178.87

The R^2 of 0.9112 on the log scale indicates the model explains 91% of variance in log-transformed CLV. The MAE of $1,389 on an average CLV of $7,934 represents ~17.5% mean absolute percentage error - actionable for marketing and retention decisions.

8.2 Feature Importance

Figure 6 shows the top 15 feature importances from the Tuned Random Forest. Two features account for over 90% of the model's predictive signal.

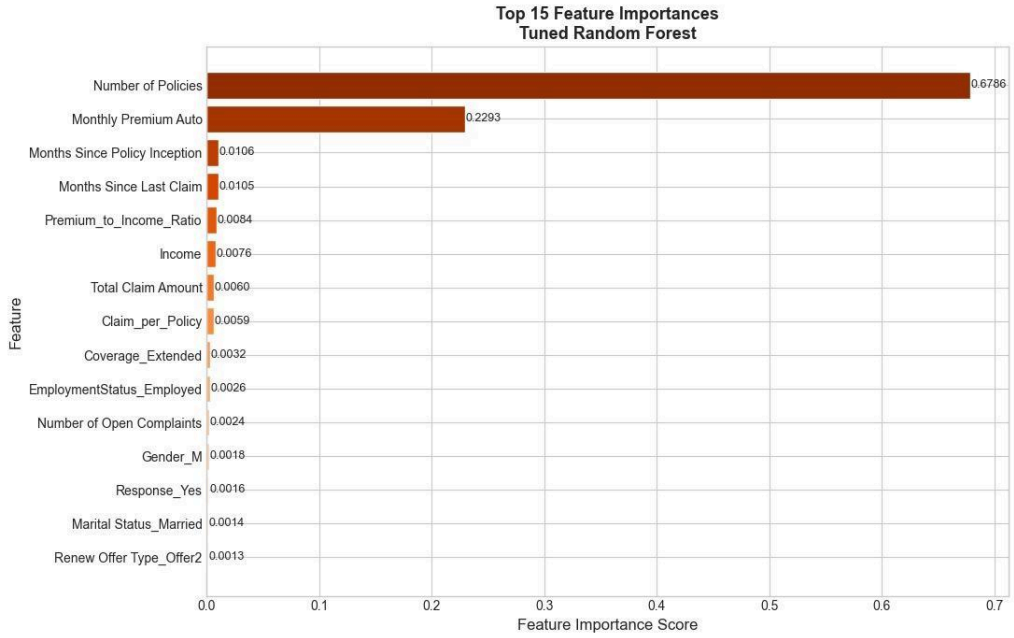


Figure 6: Top 15 Feature Importances - Tuned Random Forest. Number of Policies (0.6786) and Monthly Premium Auto (0.2293) together account for 90.8% of predictive signal.

Rank	Feature	Importance	Business Interpretation
1	Number of Policies	0.6786	Multi-policy customers generate compounding revenue - dominant CLV driver
2	Monthly Premium Auto	0.2293	Direct revenue determinant - higher premium = faster CLV accumulation
3	Months Since Policy Inception	0.0106	Longer tenure slightly associated with higher CLV
4	Months Since Last Claim	0.0105	Claims recency provides marginal risk profile signal
5	Premium_to_Income_Ratio	0.0084	Engineered feature - ranks above many raw features, validating FE effort

The engineered features Premium_to_Income_Ratio (rank 5) and Claim_per_Policy (rank 8) both outperform several raw features, validating the domain-driven feature engineering effort.

8.3 Actual vs. Predicted CLV

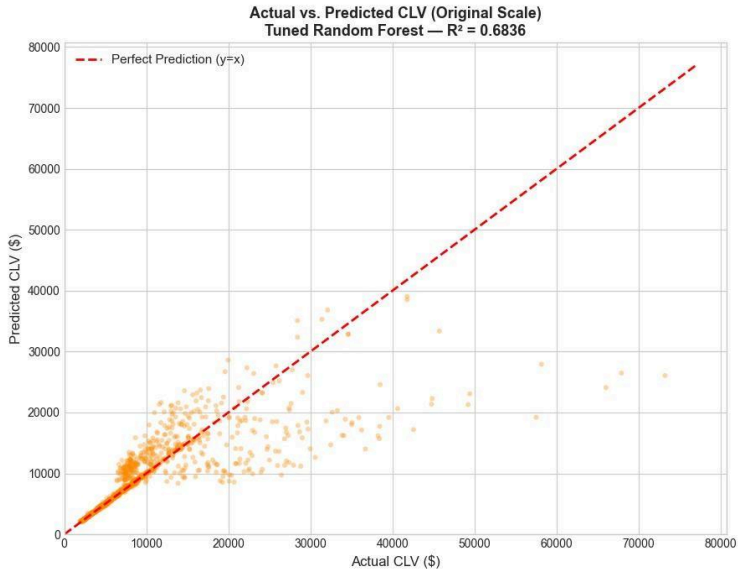


Figure 7: Actual vs. Predicted CLV (original dollar scale) - Tuned Random Forest, $R^2 = 0.6836$. Points near the red y=x line represent accurate predictions. Systematic under-prediction for very high-CLV outliers ($40,000+) reflects their sparsity in training data.

Strong alignment along the y=x line holds below $20,000. Under-prediction for $40,000+ outliers is expected given their rarity in training data.

8.4 Residuals Analysis

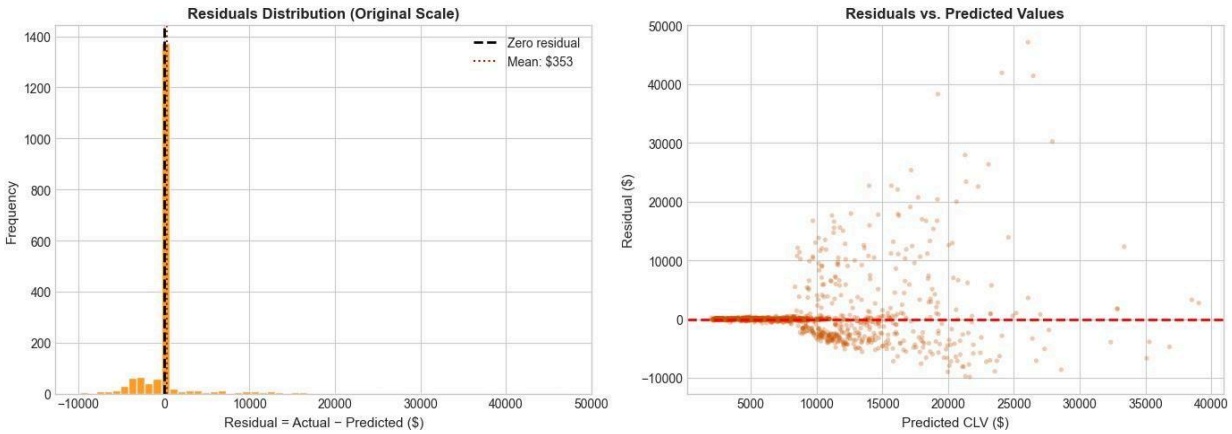


Figure 8: Residuals Distribution (left) and Residuals vs. Predicted CLV (right). The distribution is sharply peaked around zero (mean +$353). Heteroscedasticity - growing error spread with CLV magnitude - is expected after back-transforming log-scale predictions.

The mean residual is +$353.30 (slight under-prediction bias), std dev $4,022.50, and skewness 4.84 driven by sparse high-CLV under-predictions. For the ~70% of customers below $10,000, residuals are tightly clustered around zero, confirming reliable predictions across the majority of the portfolio.

8.5 Customer Segment Accuracy

Figure 9 evaluates prediction accuracy within each CLV tier using a 20% error margin threshold - a standard business-actionability benchmark.

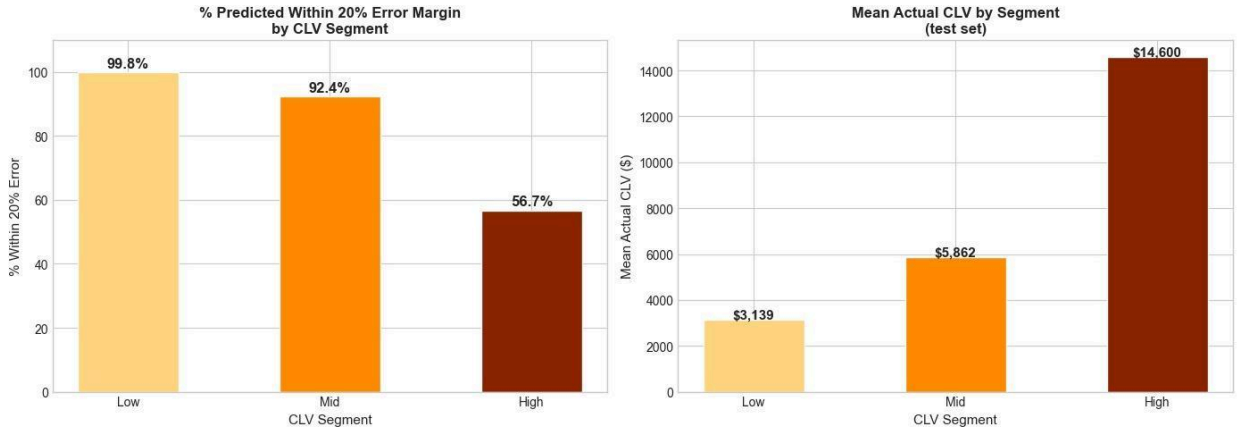


Figure 9: Segment Accuracy - % predicted within 20% error margin by CLV tier (left) and mean actual CLV by segment (right). Low and Mid segments are predicted with near-perfect accuracy; High-CLV customers remain the most challenging.

Segment	CLV Threshold	Mean CLV	% Within 20% Error	Confidence Level
Low	≤ $4,530	$3,139	99.8%	Extremely high - fully model-driven decisions appropriate
Mid	$4,530–$7,715	$5,862	92.4%	High - automated campaigns can be model-driven
High	> $7,715	$14,600	56.7%	Moderate - use as input, not sole decision basis

8.6 Strengths and Limitations

Key strengths: tree ensembles capture non-linear interactions (R^2 0.9112 vs. 0.4291 for Linear Regression), log1p transformation reduces outlier influence, domain-engineered features outrank several raw inputs, and Pipeline-based CV prevents data leakage. Key limitations: the dataset is a cross-sectional snapshot rather than longitudinal data; 25.4% zero-income customers have heterogeneous CLV trajectories; the model under-predicts the sparse $40,000+ tail; and the absence of churn labels means a full Expected CLV metric (CLV × (1−P(Churn))) cannot be computed.

8.8 Real-World Applications

CLV scores computed 60–90 days before expiry stratify renewal outreach by tier (agent for High, digital for Mid, automated for Low). At acquisition, prioritize Luxury SUV/Car owners (mean CLV $17,000+) with multi-policy bundling incentives - adding a second policy at inception has a disproportionate CLV impact per the feature importance findings. Single-policy Mid-CLV customers with above-median income are the optimal cross-sell target.

8.9 Suggested Future Improvements

Priority improvements: (1) integrate longitudinal claims sequences via LSTM or TCN for behavioral trajectory signals; (2) multiply CLV scores by a churn probability model to produce Expected CLV; (3) add SHAP

instance-level explanations for agent-facing deployments; and (4) implement quarterly retraining with automated drift detection.

9. Conclusion

This project developed a complete machine learning pipeline for predicting Customer Lifetime Value in auto insurance. Using 9,134 customer records and 24 features, five regression models were trained, cross-validated, and hyperparameter-tuned. The Tuned Random Forest achieved the strongest performance: Test $R^2 = 0.9112$ (log scale), RMSE = $4,038, and MAE = $1,389 (original dollar scale).

Number of Policies (0.68) and Monthly Premium Auto (0.23) together account for 90%+ of model signal, directly implying cross-selling additional policies is the highest-ROI lever for portfolio CLV growth.

Low and Mid CLV customers (~70% of the portfolio) are predicted with 99.8%/92.4% accuracy within 20% error, enabling fully automated retention campaigns. High-CLV predictions (56.7%) warrant human review. Future priorities: longitudinal data, churn integration, and SHAP explanations for agent-facing deployment.

10. References

IBM Watson Analytics (2019). Marketing Customer Value Analysis Dataset. Retrieved from https://www.kaggle.com/datasets/pankajjsh06/ibm-watson-marketing-customer-value-data